Goldbach-type sums $a + b = N$ with $\Omega(ab) \leq 32$

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Abstract

We prove that every integer $N \geq 2$ can be written as $N = a + b$ with positive integers a, b such that $\Omega(ab) \leq 32$, where Ω counts prime factors with multiplicity. This improves the bound $\Omega(ab) \leq 33$ of Campbell. The argument stays inside Campbell's framework, the explicit Friedlander–Iwaniec lower-bound sieve applied to the products $n(N-n)$. Three elementary reductions lead to a sieve estimate at a level depending on the parity of N; the hard case is odd N, with $z = (N^2/8)^{1/32}$. We close it with a per-interval dimension constant, per-interval sieve parameters and pre-sieving by both 3 and 5. The intermediate range is certified in ball arithmetic over 117 415 intervals, and the large range by a monotonicity argument; the even case is certified in the same way. For completeness we include a self-contained proof of the explicit Friedlander–Iwaniec lower bound with the dimension condition imposed only at the sifting level.

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1. Introduction

Let $\Omega(n)$ denote the number of prime factors of n counted with multiplicity. Chen's theorem [1] represents every sufficiently large even integer as a prime plus an integer with at most two prime factors, and Rényi's theorem [2] gives a uniform bound for all even $N \geq 4$ with an unspecified constant. Johnston and Starichkova [3] made Rényi's theorem explicit: every even $N \geq 4$ is $p + r$ with p prime and $\Omega(r) \leq 395$, and $\Omega(r) \leq 31$ under the generalised Riemann hypothesis. Their proof uses the explicit linear sieve of Bordignon, Johnston and Starichkova [4].

Relaxing the condition that one summand be prime, one asks for $N = a + b$ with $\Omega(ab)$ small. Dudek and Dunn [5] proved $\Omega(ab) \leq 40$ for every $N \geq 2$, applying the explicit Friedlander–Iwaniec $\Lambda^-\Lambda^2$ lower-bound sieve [6]. In their setting the extremal case $w = 3$ of the dimension condition forces the constant $K = 3$. Campbell [7] removed this case by pre-sieving at the prime 3 and obtained $\Omega(ab) \leq 33$. He also wrote that $\Omega(ab) \leq 32$ "appears to be within reach" and listed four possible refinements: K computed per interval; s and δ optimised per interval; pre-sieving by 3 and 5; sharper constants [7, §6].

Theorem 1.1. *Every integer $N \geq 2$ can be written as $N = a + b$ with a, b positive integers and*

$$\Omega(ab) \leq 32.$$

Campbell applies Theorem 2.2 with the dimension condition read at the sifting level only ("for each value of z under consideration" [7, (2.1)]); the global reading, over all pairs $2 \leq w < z'$, would force $K \geq 2$, which is incompatible with his constant $K = 1.146$ (Remark 4.2). For completeness we prove that this local reading suffices (Theorem 4.1, Appendix Appendix A). We do not claim novelty for this statement: the proof is Selberg's

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$\Lambda^-\Lambda^2$ argument as presented by Blight [8], the dimension condition enters only through a Rankin bound at the level z, and the statement may well be implicit in [6, §7]; we include it so that the paper does not depend on how the condition is read.

Where 33 *comes from.* The bound 33 comes only from odd N. With $z = N^{1/r}$ and $a_n < N^2$, [7, Lemma 5.2] gives $\Omega(ab) < 2r$ for even N and $\Omega(ab) < 2r + 1$ for odd N; with $r = 16.5$ this is ≤ 32 and ≤ 33 respectively. Three elementary observations (Section 3) reduce the problem to an estimate at a level depending on the parity of N: for even N the sifting level $z = (N^2/4)^{1/33}$ suffices, and this case has a wide margin; the prime-gap lemma [7, Lemma 5.1] holds under $1854m < 2^{M-1}$ instead of $1854m \le 2^{M-2}$, which covers $N \le N_0$ with $M = 32$; and for odd N one has $a_n \le (N^2 - 1)/8$, so the level $z = (N^2/8)^{1/32}$, i.e. $N = \sqrt{8}\, z^{16}$, suffices. The odd case at this level is essentially Campbell's "$r = 16$" with an extra factor $\sqrt{8}$ in the main term. With his uniform constants, Campbell estimated that the analytic range would then start only at N of order 10^{192}, with a finite range up to z of order 10^{12} [7, §6]. We close it with the first three refinements listed above; no new sieve theory is needed. The per-interval constant is essential (Section 6.2).

Organisation. Section 2 recalls Campbell's framework, Section 3 proves the reductions, Section 4 gives the sieve input and the pre-sieving by $\{3, 5\}$, and Section 5 proves Theorem 1.1. Section 6 explains why the method does not give $\Omega(ab) \le 31$.

2. Notation and Campbell's framework

Throughout, p and q denote primes, $\omega(n)$ is the number of distinct prime factors of n, and $\tau_4(n) = \#\{(d_1, d_2, d_3, d_4) \in \mathbb{Z}_{>0}^4 : d_1 d_2 d_3 d_4 = n\}$. We follow [7, §§2–3].

2.1. The sieve

Let $\mathcal{A} = (a_n)_{n \in I}$ be a finite indexed sequence of positive integers and $\mathcal{P}$ a set of sifting primes. For square-free d composed of primes of $\mathcal{P}$ put $\mathcal{A}_d = (a_n)_{n \in I,\, d \mid a_n}$ and suppose

$$|\mathcal{A}_d| = X g(d) + r_d,$$

where $X > 0$ and g is multiplicative with $0 < g(p) < 1$ for $p \in \mathcal{P}$. For $z \ge 2$ let $P(z) = \prod_{p<z,\, p \in \mathcal{P}} p$, $S(\mathcal{A}, z) = \#\{n \in I : (a_n, P(z)) = 1\}$ and $V(z) = \prod_{p \mid P(z)} (1 - g(p))$.

Definition 2.1 ([7, (2.1)])**.** Let $\kappa \ge 1$ and $K \ge 1$. The function g satisfies the dimension condition with parameters κ and K if, for each value of z under consideration and every $2 \le w < z$,

$$\prod_{\substack{w \le p < z \\ p \in \mathcal{P}}} (1 - g(p))^{-1} \le K \Big(\frac{\log z}{\log w} \Big)^{\kappa}. \tag{2.1}$$

Theorem 2.2 ([7, Theorem 2.1]; from [6, Theorems 7.7 and 7.1])**.** *With the notation above, let $D \ge z$, $s = \log D / \log z$ and $k = \kappa + \log K$. If $s \ge 2k + 3$, then*

$$S(\mathcal{A}, z) \ge X V(z) F(s, k) - 2 R_4(\mathcal{A}, D),$$

where

$$F(s, k) = 1 - \frac{s+3}{2e^k} \Big(\frac{2ek}{s-3} \Big)^{(s-3)/2}, \qquad R_4(\mathcal{A}, D) = \sum_{\substack{d \mid P(z) \\ d < D}} \tau_4(d)\, |r_d|. \tag{2.2}$$

We do not use Theorem 2.2 as a black box: all applications go through Theorem 4.1 below, which is proved in Appendix Appendix A and requires the dimension condition only at the level z.

2.2. The sequence

Fix $N \geq 2$. For $1 \leq n \leq N-1$ let $a_n = n(N-n)$ if N is even and $a_n = n(N-n)/2$ if N is odd, let $\mathcal{B} = (a_n)_{1 \leq n \leq N-1}$ and $Y := N-1$ [7, §3]. The local density is the multiplicative function with

$$g(2) = \tfrac{1}{2}, \qquad g(p) = \tfrac{2}{p} \ (p \nmid N), \qquad g(p) = \tfrac{1}{p} \ (p \mid N), \qquad p \text{ odd},$$

and for square-free ℓ the remainder ρ_ℓ of the auxiliary sequence is defined by

$$|\mathcal{B}_\ell| = Y g(\ell) + \rho_\ell, \qquad \rho_1 = 0, \tag{2.3}$$

which is [7, (3.1)]. We write $g^*(2) = \frac{1}{2}$ and $g^*(p) = 2/p$ for odd p; then $g \leq g^*$ pointwise, uniformly in N.

Lemma 2.3 ([7, proof of Lemma 4.3], via [9, Theorem 7.2])**.** *For every square-free ℓ, $|\rho_\ell| \leq 2^{\omega(\ell)}$.*

2.3. Pre-sieving

Let $Q = \{3\}$ (Campbell) or $Q = \{3, 5\}$ (this paper). Put

$$I_Q = \{1 \leq n \leq N-1 : \ q \nmid a_n \text{ for all } q \in Q\}, \qquad \mathcal{A} = (a_n)_{n \in I_Q},$$

$X = |I_Q|$ and $\mathcal{P} = \{p : \ p \notin Q\}$. For $Q = \{3\}$, Campbell shows $|\mathcal{A}_d| = Xg(d) + r_d$ with $r_d = \rho_d - \rho_{3d} + g(d)\rho_3$ [7, (3.3)], the bound $|r_d| \leq \frac{19}{6} 2^{\omega(d)}$ and $X \geq (N-1)/3$. The case $Q = \{3, 5\}$ is treated in Section 4.2. In both cases we write C_Q for a constant with $|r_d| \leq C_Q 2^{\omega(d)}$ for $d \mid P(z)$, $d > 1$; thus $C_{\{3\}} = 19/6$ and $C_{\{3,5\}} = 11$.

Since $d \mid P(z)$ is square-free, $\tau_4(d) 2^{\omega(d)} = \tau_8(d)$, and Rankin's trick gives, for $0 < \delta < 1$ [7, (4.4)],

$$R_4(\mathcal{A}, D) \leq C_Q \, D^\delta \prod_{\substack{p < z \\ p \in \mathcal{P}}} \Big(1 + \frac{8}{p^\delta}\Big). \tag{2.4}$$

2.4. Campbell's proof

Campbell sets $N_0 = 11746518036554764849893443 9$, $z = N^{1/16.5}$, $z_0 = N_0^{1/16.5} = 38.018\ldots$, $z_1 = 10^8$ and $N_1 = 10^{132}$, and splits the proof into three ranges [7, §§4–5]: Case 1, $2 \leq N \leq N_0$, by prime-gap data ([7, Lemma 5.1]); Case 2, $z_0 < z < z_1$, by a floating-point finite verification with $K = 1.146$, $s = 14.8$ and $\delta \in \{0.20, 0.60, 0.75, 0.85, 0.93\}$; Case 3, $N \geq N_1$, analytically with $K = 1.097$, $s = 14.66$ and $F > 0.0563$. Cases 2 and 3 establish $S(\mathcal{A}, z) > 0$, and [7, Lemma 5.2] converts this into the bound on $\Omega(ab)$.

2.5. Inputs taken from the literature

Besides Theorem 4.1 (proved here) and Lemma 2.3, the proof uses:

(H3) Rosser–Schoenfeld [10, Theorem 5]: $\sum_{p \leq x} 1/p > \log\log x + B - \frac{1}{2\log^2 x}$ for $x > 1$, and $\sum_{p \leq x} 1/p < \log\log x + B + \frac{1}{2\log^2 x}$ for $x \geq 286$.

(H4) $\sum_{p \leq x} \frac{\log p}{p} < \log x$, as in [7, §4] ("corollary to Theorem 6" of [10]).

(H5) The Meissel–Mertens constant satisfies $B = 0.26149721284764\ldots < 0.2614972128477$.

(H7) Every prime $q < G := 1014123199963633 10923$ is followed by a prime q' with $q' - q \leq 1854$ [11], as used in [7, §5].

3. Reductions

3.1. The prime-gap range

Lemma 3.1. *Let m be 1 or a prime and M a positive integer with $1854m < 2^{M-1}$. Then for every $2 \le N \le Gm$ there are positive integers a, b with $N = a + b$ and $\Omega(ab) \le M$.*

Proof. Every prime factor is at least 2, so $2^{\Omega(x)} \le x$ for every integer $x \ge 1$. Hence if $1 \le x \le 1854m < 2^{M-1}$, then $\Omega(x) \le \log_2 x < M - 1$, and since $\Omega(x)$ is an integer, $\Omega(x) \le M - 2$.

If $N \le 2m$, take $a = 1$, $b = N - 1 \ge 1$. Then $b < 2m \le 1854m$, so $\Omega(ab) = \Omega(N - 1) \le M - 2 < M$.

If $N > 2m$, let p be the largest prime with $p < N/m$ (it exists since $N/m > 2$) and p' the next prime, so that $p < N/m \le p'$. Since $p < N/m \le G$, (H7) gives $p' - p \le 1854$. Hence $0 < N - mp \le m(p' - p) \le 1854m$ and $\Omega(N - mp) \le M - 2$. With $a = mp$ and $b = N - mp$ we have $\Omega(mp) \le 2$, so $\Omega(ab) \le M$. □

Corollary 3.2. *For every $2 \le N \le N_0$ there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

Proof. Take $M = 32$ and the prime $m = 1158293$. Then

$$1854m = 2147475222 < 2^{31} = 2147483648, \qquad Gm = N_0,$$

and Lemma 3.1 applies. □

Remark 3.3. [7, Case 1] writes $1854m = 2147475222 < 2^{31} = 2^{M-2}$ with $M = 33$. The gain is only the rounding of $\log_2$ (the strict inequality for $N \le 2m$ is already in [7]). The slack is necessary: $1854m \le 2^{30} = 1073741824$ is false, so the hypothesis of [7, Lemma 5.1] cannot be used with $M = 32$ and this m.

3.2. Even N

Lemma 3.4. *Let $Q \in \{\{3\}, \{3, 5\}\}$, let $N > N_0$ be even and let $z := (N^2/4)^{1/33}$, i.e. $N = 2z^{16.5}$. If $S(\mathcal{A}, z) > 0$, then there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

Proof. Since $N > N_0$ we have $z > (N_0^2/4)^{1/33} > 36$, so $z > 5$. As $S(\mathcal{A}, z) > 0$ there is $n \in I_Q$ with $(a_n, P(z)) = 1$; since no $q \in Q$ divides a_n and $\mathcal{P}$ contains every other prime, $a_n = n(N - n)$ has no prime factor less than z. Let $t = \Omega(a_n)$. Since $a_n \le N^2/4 = z^{33}$, we have $z^t \le a_n \le z^{33}$, so $t \le 33$, and $t = 33$ would force $a_n = z^{33}$ with every prime factor equal to z; then z is a prime and $(N/2)^2 = z^{33}$ is a square, which is impossible since 33 is odd. Hence $t \le 32$, and with $a = n$, $b = N - n$ we have $\Omega(ab) = t \le 32$. □

Remark 3.5. Campbell's own proof gives $\Omega(ab) \le 32$ for even N already, through [7, Lemma 5.2] with $r = 16.5$; its finite range, however, is verified in floating point with a cushion of 0.10 in the logarithm (reproducing the scripts of [12], tag `arxiv-v1`, gives 227 intervals and minimal log-ratio 0.1000056). We do not use it: the even case is certified here at the level of Lemma 3.4 (Proposition 5.2 and Proposition 5.5), with margins at least 5.02.

3.3. Odd N

Lemma 3.6. *If N is odd and $1 \le n \le N - 1$, then $a_n = n(N - n)/2 \le (N^2 - 1)/8 < N^2/8$.*

Proof. The function $n \mapsto n(N - n)$ is concave and symmetric about $N/2$. Over the integers its maximum is at $n = (N \pm 1)/2$, where it equals $\frac{N-1}{2} \cdot \frac{N+1}{2} = \frac{N^2-1}{4}$. □

Lemma 3.7. *Let $Q \in \{\{3\}, \{3, 5\}\}$, let $N > N_0$ be odd and let $z := (N^2/8)^{1/32}$, i.e. $N = \sqrt{8}\, z^{16}$. If $S(\mathcal{A}, z) > 0$, then there are $a, b \ge 1$ with $N = a + b$ and $\Omega(ab) \le 32$.*

Proof. Since $N > N_0$ we have $z > z_0^* := (N_0^2/8)^{1/32} = 39.91606854\ldots$; in particular $z > 5$. As $S(\mathcal{A}, z) > 0$ there is $n \in I_Q$ with $(a_n, P(z)) = 1$. Since no $q \in Q$ divides a_n and $\mathcal{P}$ contains every other prime, a_n has no prime factor less than z. Let $t = \Omega(a_n)$. By Lemma 3.6, $z^t \le a_n < N^2/8 = z^{32}$, so $t \le 31$. With $a = n$, $b = N - n$ we have $ab = 2a_n$ and $\Omega(ab) = t + 1 \le 32$. □

The level z depends on N, and no lemma of [7] uses a fixed exponent r except [7, Lemma 5.2]. The bound $X \geq (N-1)/3$ continues to hold, and $z_0^* > z_0$, so the ranges of [7, Lemma 4.1] remain covered.

In the computation we start from the floating-point number

$$u_0 = 39.91606854834768,$$

for which $8u_0^{32} \leq N_0^2$ is checked in exact rational arithmetic; hence every $N > N_0$ has $z > u_0$.

For even N the same computation starts from $u_0^{\mathrm{e}} = 36.454424943231$, for which $4(u_0^{\mathrm{e}})^{33} \leq N_0^2$ is checked in exact rational arithmetic.

Corollary 3.8. *Theorem 1.1 follows from Corollary 3.2 and the statement*

$$S(\mathcal{A}, z) > 0 \quad \textit{for every odd } N \textit{ with } z = (N^2/8)^{1/32} > u_0, \textit{ and every even } N \textit{ with } z = (N^2/4)^{1/33} > u_0^{\mathrm{e}}.$$

We prove it with $Q = \{3, 5\}$. All bounds depend on N only through z and the parity of N, so the proof is uniform in N of each parity. We write $N = c_\pi z^{A_\pi}$ with $(c_\pi, A_\pi) = (\sqrt{8}, 16)$ for odd N and $(2, 16.5)$ for even N.

4. The sieve input

4.1. The dimension condition at the sifting level

Theorem 4.1 (the lower bound at a fixed level)**.** *Let $\mathcal{A}$, $\mathcal{P}$, g and r_d be as in Section 2, fix $z \geq 2$, and let $\kappa > 0$ and $K \geq 1$ be such that*

$$\prod_{\substack{w \leq p < z \\ p \in \mathcal{P}}} (1 - g(p))^{-1} \leq K\Big(\frac{\log z}{\log w}\Big)^{\kappa} \qquad \textit{for every } 2 \leq w < z, \tag{4.1}$$

at this z only. Let $D \geq z$, $s = \log D/\log z$ and $k = \kappa + \log K$. If $s \geq 2k + 3$, then $S(\mathcal{A}, z) \geq XV(z)F(s, k) - 2R_4(\mathcal{A}, D)$.

Theorem 4.1 is Theorem 2.2 with the dimension condition required at the sifting level only, which is Definition 2.1 as written in [7, (2.1)]; it is proved in Appendix Appendix A. Campbell's Lemma 4.1(1) is stated "for every $z \geq z_0$ and every $2 \leq w < z$", and he explicitly uses "the restrictions $z \geq z_0$ ... rather than estimates uniform over a larger range" [7, §4]; this is the form of the statement that such a use needs. From now on all applications of the sieve are through Theorem 4.1 with $\kappa = 2$.

Remark 4.2 (the global reading). Suppose (2.1) were required for all pairs $2 \leq w < z'$, for every z'. With $w = 2$ and $z' \to 2^+$ the product equals $(1 - g(2))^{-1} = 2$, while $(\log z'/\log 2)^2 \to 1$; this forces $K \geq 2 > 1.146$. (With $w = 5$ and $z' \to 5^+$ one obtains $K \geq 5/3$.) So the constants of [7] are compatible only with a reading that is not global. A hypothetical reading "for all $z' \in [z_0, z]$" would save [7] but not the present argument, since it amounts to a uniform constant, which fails here (Section 6.2). Theorem 4.1 makes the question moot.

Let $h(p) = g(p)/(1 - g(p))$ for $p \in \mathcal{P}$, extended multiplicatively to square-free integers composed of primes of $\mathcal{P}$, and for $U \geq 1$ put $I(U, z) = \sum_{d \mid P(z),\, d \geq U} h(d)$; note that $V(z)^{-1} = \sum_{d \mid P(z)} h(d)$. Proposition 4.4 gives the Rankin bound for $I(U, z)$ with general K, using the dimension condition only at the level z; it is the analytic input of Theorem 4.1.

Remark 4.3. The argument of Appendix Appendix A is Selberg's $\Lambda^-\Lambda^2$ construction as presented in Blight's thesis [8, §§2.2–2.3] (advisor H. Iwaniec), where the dimension condition enters only through the bound for $I(U, z)V(z)$ quoted from [6, p. 111]. We replace that quotation by Proposition 4.4, which uses the condition at the level z only.

Proposition 4.4. *Fix z, let $\kappa > 0$ and $K \geq 1$, and let g satisfy $0 < g(p) < 1$ for $p \in \mathcal{P}$. Put $\mathcal{L}_z(t) = \sum_{t \leq p < z,\, p \in \mathcal{P}} -\log(1 - g(p))$ and suppose*

$$\mathcal{L}_z(t) \leq \kappa \log \frac{\log z}{\log t} + \log K \qquad (2 \leq t < z).$$

Let $k = \kappa + \log K$ and $u = \log U / \log z$. If $u \geq k$, then

$$I(U, z)V(z) \leq e^{-k}\Big(\frac{ek}{u}\Big)^u.$$

With $u = t/2$ this is $e^{-k}(2ek/t)^{t/2}$; with $u = (s-3)/2$ it gives $F(s, k)$, including the condition $s \geq 2k + 3$.

Proof. If $u = k$ the right-hand side equals 1, and $I(U, z) \leq V(z)^{-1}$; so assume $u > k$. For $\varepsilon > 0$, $I(U, z) \leq U^{-\varepsilon} \prod_{p<z}(1 + h(p)p^{\varepsilon})$ and $V(z)^{-1} = \prod_{p<z}(1 + h(p))$. Since $(1 + hp^{\varepsilon})/(1 + h) = 1 + g(p)(p^{\varepsilon} - 1)$,

$$I(U, z)V(z) \leq U^{-\varepsilon} \prod_{p<z} \big(1 + g(p)(p^{\varepsilon} - 1)\big) \leq \exp\Big(-\varepsilon \log U + \sum_{p<z} g(p)(p^{\varepsilon} - 1)\Big).$$

Take $\varepsilon = c/\log z$ with $c > 0$ and $f(t) = t^{\varepsilon} - 1$, which is non-negative and increasing. Since $g \leq -\log(1 - g)$, partial summation gives $\sum_{p<z} g(p)f(p) \leq f(2)\mathcal{L}_z(2) + \int_2^z f'(t)\mathcal{L}_z(t)\, dt$. Insert the hypothesis. The $\log K$ terms give $\log K\, (f(2) + \int_2^z f') = \log K\, (e^c - 1)$. With $\varphi(t) = \log(\log z/\log t)$, $\varphi(z) = 0$, integration by parts turns the κ terms into

$$\kappa \int_2^z \frac{t^{\varepsilon} - 1}{t \log t}\, dt = \kappa \int_{\varepsilon \log 2}^{c} \frac{e^v - 1}{v}\, dv \leq \kappa(e^c - 1),$$

because $(e^v - 1)/v \leq e^v$. The total is at most $k(e^c - 1)$, so $I(U, z)V(z) \leq \exp(-cu + k(e^c - 1))$. Choosing $e^c = u/k > 1$ gives the claim. □

4.2. Pre-sieving by $Q = \{3, 5\}$

From now on $Q = \{3, 5\}$, $I_{15} := I_Q$ and $\mathcal{P} = \{p : p \notin \{3, 5\}\}$.

Lemma 4.5. *For square-free d composed of primes of $\mathcal{P}$,*

$$|\mathcal{A}_d| = Xg(d) + r_d, \qquad r_d = \sum_{e \mid 15} \mu(e)\big(\rho_{de} - g(d)\rho_e\big), \qquad r_1 = 0.$$

For $Q = \{3\}$ the same computation gives $r_d = \rho_d - \rho_{3d} + g(d)\rho_3$, which is [7, (3.3)].

Proof. Since $(d, 15) = 1$, $de \mid a_n$ if and only if $d \mid a_n$ and $e \mid a_n$, so inclusion–exclusion gives $|\mathcal{A}_d| = \sum_{e \mid 15} \mu(e)|\mathcal{B}_{de}|$. By (2.3) and multiplicativity, $\sum_e \mu(e)|\mathcal{B}_{de}| = Yg(d)(1 - g(3))(1 - g(5)) + \sum_e \mu(e)\rho_{de}$. Likewise $X = \sum_e \mu(e)|\mathcal{B}_e| = Y(1 - g(3))(1 - g(5)) + \sum_e \mu(e)\rho_e$. Eliminating $Y(1 - g(3))(1 - g(5))$ gives the identity. □

Lemma 4.6. *For $d \mid P(z)$ with $d > 1$, $|r_d| \leq 11 \cdot 2^{\omega(d)}$.*

Proof. By Lemma 2.3, $|\rho_3|, |\rho_5| \leq 2$ and $|\rho_{15}| \leq 4$. For $d > 1$ composed of primes of $\mathcal{P}$ we have $g(d) \leq \frac{1}{2}$ and $2^{\omega(d)} \geq 2$. Hence, using $\rho_1 = 0$,

$$|r_d| \leq \sum_{e \mid 15} 2^{\omega(de)} + \tfrac{1}{2}\big(|\rho_3| + |\rho_5| + |\rho_{15}|\big) \leq 9 \cdot 2^{\omega(d)} + 4 \leq 11 \cdot 2^{\omega(d)},$$

where $9 = 1 + 2 + 2 + 4$. □

Remark 4.7. The analogue in [7] is $3 \cdot 2^{\omega} + \frac{1}{3} \le \frac{19}{6} 2^{\omega}$. Since $\max(|\rho_3| + |\rho_5| + |\rho_{15}|) = 10/3$ (periodic modulo 60), one could use $(9 + \frac{5}{6})2^{\omega(d)}$, i.e. $C = 9.83$, which would add about $\log(11/9.83) \approx 0.11$ to the margins. This finer value is not needed: the margins are positive with $C_Q = 11$, which is the value used throughout and in the certificate.

Lemma 4.8. $X \ge (N-1)/5 - 3 = (N-16)/5$.

Proof. By the Chinese remainder theorem, the admissible classes modulo 15 are products of admissible classes modulo 3 and modulo 5. Modulo 3, $3 \mid a_n$ if and only if $n \equiv 0$ or $n \equiv N$, so there is 1 admissible class if $3 \nmid N$ and 2 if $3 \mid N$; modulo 5 there are 3 or 4. Hence there are at least 3 admissible classes modulo 15, each containing at least $\lfloor (N-1)/15 \rfloor > (N-1)/15 - 1$ elements of $[1, N-1]$. Therefore $X > 3\big((N-1)/15 - 1\big) = (N-1)/5 - 3$. □

Theorem 4.1 applies without change, since it is stated for an arbitrary sequence and set of sifting primes, and $0 < g(p) < 1$ on $\mathcal{P}$. Moreover,

$$V(z) \ge \tfrac{1}{2} \prod_{7 \le p < z} \Big(1 - \frac{2}{p}\Big), \tag{4.2}$$

the factor $1 - g(5)$ having moved from V into X, and (2.4) holds with $C_Q = 11$. In Lemma 3.7 we used $z > 5$.

Remark 4.9. Removing 5 from $\mathcal{P}$ removes the worst case of [7, Lemma 4.1] ($z \to 43^+$, $w = 5$, constant 1.1457). The new worst case is $z \to 43^+$, $w = 11$, with $K = 1.089970$ on the cell $(42.9989, 43.0032]$. Campbell anticipated this: "the prime 5, which is responsible for the largest values of K" [7, §6].

4.3. The dimension constant on an interval

Lemma 4.10. *Let $I = (u, v]$ with $u \ge u_0^{\mathrm{e}}$ (note $u_0^{\mathrm{e}} < u_0$) and let*

$$K_I \;\ge\; \sup_{z \in I} \sup_{2 \le w < z} \prod_{\substack{w \le p < z \\ p \in \mathcal{P}}} (1 - g^*(p))^{-1} \Big(\frac{\log w}{\log z}\Big)^2 .$$

Then for every N and every $z \in I$, the condition of Theorem 4.1 holds at z with $\kappa = 2$ and $K = K_I$, and $K_I \ge 1$. Moreover, the supremum is approached as $z \to u^+$ or $z \to q^+$ with $q \in I \cap \mathcal{P}$, and it suffices to take $w = 2$ or $w \in \mathcal{P}$.

Proof. The product increases with g, and $g \le g^*$, so the bound is uniform in N. Letting $w \to z^-$ shows $K_I \ge 1$. For a fixed set of included primes the quantity decreases in z, so on each interval between consecutive primes of $\mathcal{P}$ its largest value is approached at the left end; as a function of w the product is constant between consecutive primes while $(\log w)^2$ increases. This is the argument of Case 3 of [7, Lemma 4.1]. □

Since K_I satisfies (4.1) at every $z \in I$, Theorem 4.1 applies at each $z \in I$ with $k_I = 2 + \log K_I$; no monotonicity of F in k is needed.

For the analytic range let Z be an even integer, put

$$T := \frac{1}{Z-2} + \frac{1}{Z-1} \;\ge\; \sum_{n \ge Z} \frac{2}{n(n-2)}$$

(telescoping: $\frac{2}{n(n-2)} = \frac{1}{n-2} - \frac{1}{n}$), and

$$K_{\mathrm{ex}}(Z) = \sup_{2 \le w < Z} \prod_{\substack{w \le p < Z \\ p \in \mathcal{P}}} (1 - g^*(p))^{-1} \Big(\frac{\log w}{\log Z}\Big)^2 .$$

Lemma 4.11. *For every $z \ge Z \ge 286$ and every $2 \le w < z$, the dimension condition holds at z with $\kappa = 2$ and*

$$K_{\mathrm{an}}(Z) = \max\bigl(1, K_{\mathrm{ex}}(Z)\bigr) \exp\Bigl(\frac{2}{\log^2 Z} + T\Bigr).$$

Proof. (i) $w \ge Z$. For odd $p \in \mathcal{P}$, $-\log(1-2/p) \le \frac{2}{p} + \frac{2}{p(p-2)}$, from the Taylor series as in [7, (4.1)]. By (H3), the lower bound taken at w^- and the upper bound at z,

$$2 \sum_{w \le p < z} \frac{1}{p} \le 2 \log \frac{\log z}{\log w} + \frac{1}{\log^2 z} + \frac{1}{\log^2 w} \le 2 \log \frac{\log z}{\log w} + \frac{2}{\log^2 Z},$$

and $\sum_{p \ge Z} \frac{2}{p(p-2)} \le T$. (ii) $w < Z$. Split the product at Z. The part over $w \le p < Z$ is at most $K_{\mathrm{ex}}(Z)(\log Z/\log w)^2$, and the part over $Z \le p < z$ is case (i) with $w = Z$ (Z is even, hence not prime). Multiply. □

This is Cases 4 and 5 of [7, Lemma 4.1] with the cut at Z instead of 293, which is what makes K_{an} small.

5. Proof of the main theorem

Throughout, $Q = \{3, 5\}$, $C_Q = 11$, $\ell_2 = \log 2$, $\ell_p = -\log(1-2/p)$ for odd p, and for $x > 2$

$$L(x) = \sum_{\substack{p < x \\ p \in \mathcal{P}}} \ell_p, \qquad E_\delta(x) = \sum_{\substack{p < x \\ p \in \mathcal{P}}} \log\bigl(1 + 8p^{-\delta}\bigr).$$

Thus $V(z) \ge e^{-L(z)}$ for every N.

5.1. A margin on an interval

Lemma 5.1. *Let $I = (u, v]$ with $u \ge u_0$ ($u \ge u_0^{\mathrm{e}}$ for even N), let K_I be as in Lemma 4.10, $k_I = 2 + \log K_I$, and let $s \ge 2k_I + 3$ with $F(s, k_I) > 0$ and $0 < \delta < 1$. Put $X_{\mathrm{lo}}(u) = (c_\pi u^{A_\pi} - 1)/5 - 3$ and*

$$m_I(s, \delta) = \log X_{\mathrm{lo}}(u) - L(v) + \log F(s, k_I) - \log(2C_Q) - s\delta \log v - E_\delta(v).$$

If $m_I(s, \delta) > 0$, then $S(\mathcal{A}, z) > 0$ for every N of the given parity with $N = c_\pi z^{A_\pi}$, $z \in I$.

Proof. Let $z \in I$ and $D = z^s$. By Lemma 4.10 and Theorem 4.1, $S(\mathcal{A}, z) \ge XV(z)F(s, k_I) - 2R_4(\mathcal{A}, D)$. Each factor is bounded at the appropriate end of I:

- $X \ge X_{\mathrm{lo}}(u)$ by Lemma 4.8, since $N = c_\pi z^{A_\pi} > c_\pi u^{A_\pi}$;
- $V(z) \ge e^{-L(v)}$, since V decreases in z, the primes below z are among those below v, and $g \le g^*$;
- $2R_4(\mathcal{A}, z^s) \le 2C_Q v^{s\delta} e^{E_\delta(v)}$ by (2.4), since $D^\delta = z^{s\delta} \le v^{s\delta}$ and the product increases in z.

Hence $\log\bigl(XV(z)F(s, k_I)/2R_4(\mathcal{A}, D)\bigr) \ge m_I(s, \delta) > 0$. □

5.2. The finite range $u_0 < z \le 10^8$

Proposition 5.2 (certified computation)**.** *There is a partition of $(u_0, 10^8]$ into 117 415 contiguous cells $I = (u, v]$, with parameters (s_I, δ_I) and constants K_I as in Lemma 4.10, such that $s_I \ge 2k_I + 3$, $F(s_I, k_I) > 0$ and*

$$m_I(s_I, \delta_I) \ge 1.118347$$

for every cell. The minimum is attained on the cell $(10342.3367,\ 10343.3709]$, with $s = 14.30443$, $\delta = 107/128$ and $K = 1.002250$. For even N the same construction on $(u_0^{\mathrm{e}}, 10^8]$ gives 118 322 cells with $m_I \ge 5.024006$, attained on $(2802.7202,\ 2803.0005]$ with $s = 14.34590$, $\delta = 102/128$ and $K = 1.005873$.

Code and output are listed under the Code availability statement.

Cells. Geometric cells of ratio $1 + 10^{-4}$ from u_0 to 10^6; above 10^6, cells $(b_j, b_{j+1}]$ with integer endpoints and relative step $2 \cdot 10^{-4}$ on $[10^6, 10^7]$ and $5 \cdot 10^{-4}$ on $[10^7, 10^8]$. The code checks that consecutive cells share endpoints, that the first starts at u_0 and that the last ends exactly at 10^8.

Dimension constant. For cells below 10^6, K_I is computed from the exact candidates of Lemma 4.10 ($z \to u^+$, $z \to q^+$, w prime), one ball per prime. For block cells $z \in (b_j, b_{j+1}]$ we use

$$\log K_I \le L_{\mathrm{hi}}(b_{j+1}) - 2\log\log b_j + \max_{w < b_{j+1}} H(w), \qquad H(w) = 2\log\log w - L(w),$$

and, inside a block $[A, B)$, $H(w) \le 2\log\log B - L_{\mathrm{lo}}(A)$. A brute-force comparison at 7 points confirms that the rigorous bound is at least the exact value, with relative excess at most $1.1 \cdot 10^{-4}$.

Parameters. $\delta \in \{j/128 : 26 \le j \le 127\}$ (102 exact dyadic values); s is optimised in floating point (grid and golden-section search) and then used as an exact constant. The conditions $s \ge 2k_I + 3$ and $F(s, k_I) > 0$ are verified in ball arithmetic on every cell.

Ball arithmetic. All bounds are evaluated with python-flint 0.9.0 (Arb) at 128 bits. Conversions from balls to floating point move one ulp outwards and are checked by exact Arb comparison.

Sums over primes. For $p < 10^6$ each term is evaluated separately as a ball. For $10^6 \le p < 10^{10}$ the primes are grouped into blocks $[A, B)$ for which the number n of primes and $S_1 = \sum p$ are computed exactly (64-bit integers). For f convex on $[A, B]$,

$$n\, f(S_1/n) \le \sum_{p \in [A,B)} f(p) \le n f(A) + \frac{(f(B) - f(A))(S_1 - nA)}{B - A}$$

(Jensen and the chord). The two functions used are convex: $\ell(x) = \sum_{j \ge 1} (2/x)^j / j$ is a sum of convex functions for $x > 2$, and $e_\delta(x) = \log(1 + 8x^{-\delta})$ satisfies

$$e''_\delta(x) = \frac{8\delta x^{-\delta-2}\left[(\delta + 1) + 8x^{-\delta}\right]}{(1 + 8x^{-\delta})^2} > 0.$$

The resulting enclosures are narrow: for $Q = \{3\}$,

$$L(10^8) \in [5.604898185,\ 5.604898206],$$

and the excess in $E_\delta(10^8)$ ranges from $6 \cdot 10^{-8}$ ($\delta \approx 0.99$) to $5 \cdot 10^{-3}$ ($\delta \approx 0.20$).

Primes. The primes up to 10^{10} are generated by a segmented sieve of Eratosthenes (20 730 blocks above 10^6). The values $\pi(10^k)$ for $k = 6, \ldots, 10$ agree with the known values (e.g. $\pi(10^{10}) = 455\,052\,511$), and the counts of 60 random blocks, of the four decades, and the exact sums of 12 blocks agree with the independent program primesieve 12.17, with no mismatch.

Floating point versus certified. The floating-point margins differ from the certified ones by at most $1.8 \cdot 10^{-13}$ over all covers. Rounding is therefore not the issue; ball arithmetic makes every bound used in the proof a proven inequality, and the separate implementations below guard, to some extent, against programming errors. Table 1 lists the minimal certified margin in each decade.

Table 1: Minimal certified margin m_I per decade, $Q = \{3, 5\}$, per-cell K.

range of z	min. certified margin
$(39.9,\ 10^2]$	+15.369
$(10^2,\ 10^3]$	+2.591
$(10^3,\ 10^4]$	+1.124
$(10^4,\ 10^5]$	**+1.118**
$(10^5,\ 10^6]$	+1.829
$(10^6,\ 10^7]$	+3.414
$(10^7,\ 10^8]$	+5.510

The bottleneck is always $z \sim 1.0$–$1.6 \cdot 10^4$, i.e. odd $N \sim 10^{64}$–10^{67}. Figure 1 plots the certified margin of both chains over the whole finite range, and Figure 2 the per-cell constants K_I that make it positive.

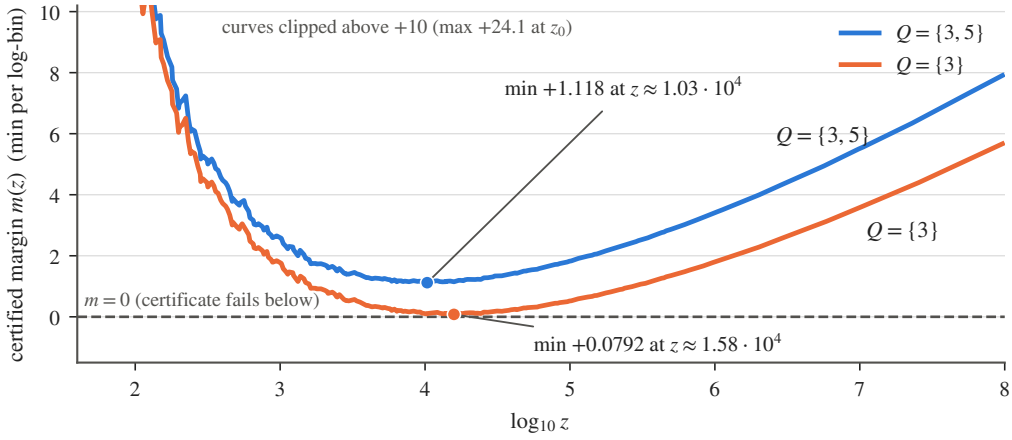


Figure 1: Certified margin m_I on the finite range, per-cell K (minimum over the cells in each of 400 logarithmic bins; ball-arithmetic lower bounds). Blue: $Q = \{3, 5\}$, 117 415 cells up to 10^8; orange: $Q = \{3\}$, same range (122 023 cells up to 10^{10} in total). Both stay above 0; the dots are the minima of Table 3 (rows I). Nothing in this figure is floating-point only.

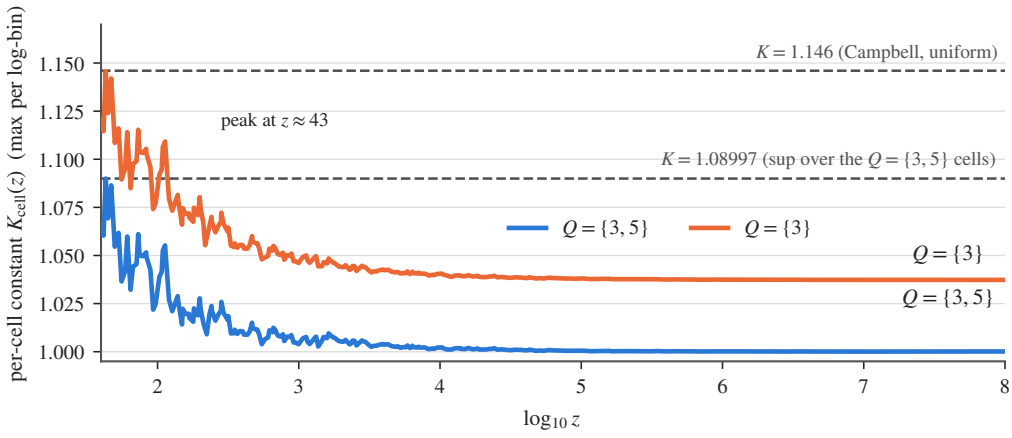


Figure 2: The certified upper bound K_I for the dimension constant in each cell (maximum per bin). The per-cell values fall below 1.01 ($Q = \{3, 5\}$) after $z \approx 10^3$ and below 1.04 ($Q = \{3\}$) after $z \approx 2 \cdot 10^4$, while a uniform constant must cover the peak near $z \approx 43$: Campbell's 1.146 (dashed) and the supremum 1.08997 for $Q = \{3, 5\}$ both make the finite range fail (Section 6.2).

Checks by separate implementations. A separate floating-point implementation, not importing the main code (147 347 geometric cells with $\varepsilon = 10^{-4}$, up to exactly 10^8), finds the minimum +1.1188 on (10342.337, 10343.371] with no failing cell. A per-prime evaluation without Jensen bounds (own sieve, mpmath) agrees with the certificate within $2 \cdot 10^{-8}$ on 7 cells when the same constant K_I is used. The identity of Lemma 4.5 was checked exactly (rational arithmetic) for 859 odd N and all $d \mid 2 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, and Lemma 3.6 by brute force for odd $N < 1500$ and 60 values of N near 30031, with no violation; these checks cover only small odd N.

5.3. The analytic range $z \geq 10^8$

Let Z be an even integer with $Z \geq 286$, $t = \log z \geq \log Z$, and $K_{\mathrm{an}} = K_{\mathrm{an}}(Z)$ from Lemma 4.11.

Lemma 5.3. *For $z \geq Z$, $V(z) \geq c_V / \log^2 z$ with*

$$c_V = e^{-L(Z)} \log^2 Z \, e^{-2/\log^2 Z - T}.$$

Proof. $V(z) \geq e^{-L(Z)} \prod_{Z \leq p < z} (1 - 2/p)$, and as in Lemma 4.11,

$$-\log \prod_{Z \leq p < z} \Big(1 - \frac{2}{p}\Big) \leq 2 \log t - 2 \log \log Z + \frac{2}{\log^2 Z} + T.$$

An upper bound L_{hi} for $L(Z)$ is used. □

Lemma 5.4. *Let $0 < \alpha < \log Z$, $\eta = \alpha/t$, $\delta = 1 - \eta$ and $D = z^s$. Then*

$$\log 2R_4(\mathcal{A}, D) \leq \log(2C_Q) + st - \alpha s + 8 \log t + 8\Big(B + \frac{1}{2t^2} - \sigma_Q\Big) + 8(e^\alpha - 1),$$

where $\sigma_{\{3,5\}} = \frac{1}{3} + \frac{1}{5}$ (and $\sigma_{\{3\}} = \frac{1}{3}$).

Proof. By (2.4) and $1 + x < e^x$, $R_4 \leq C_Q D^\delta \exp\big(8 \sum_{p<z,\, p \in \mathcal{P}} p^{-\delta}\big)$. For $p < z$ write $p^{-\delta} = p^{-1} e^x$ with $x = \eta \log p \in (0, \alpha)$. By convexity (the chord on $[0, \alpha]$, in place of $e^x < 1 + xe^x$ used in [7]), $e^x \leq 1 + x \frac{e^\alpha - 1}{\alpha}$, so $p^{-\delta} \leq \frac{1}{p} + (e^\alpha - 1) \frac{\log p}{pt}$. Summing with (H3) (upper bound) and (H4) gives $\sum_{p<z,\, p \in \mathcal{P}} p^{-\delta} \leq \log t + B + \frac{1}{2t^2} - \sigma_Q + (e^\alpha - 1)$. Finally $D^\delta = e^{st - \alpha s}$. A term $-8(e^\alpha - 1) c_Q / t$ with $c_Q = \sum_{q \in Q} \log q / q$ has been dropped; it enters the margin with a positive sign, so dropping it is conservative. □

Proposition 5.5. *Let $Q = \{3, 5\}$, $Z = 10^8$, $k = 2 + \log K_{\mathrm{an}}$, and fix s and α with $s \geq 2k + 3$ and $(A_\pi - s) \log Z > 10$. Let $M(t)$ denote the logarithm of the ratio of $XV(z)F(s,k)$ to $2R_4(\mathcal{A}, z^s)$. Then $M(t) \geq M_{\mathrm{lo}}(\log Z)$ for all $t \geq \log Z$, where*

$$M_{\mathrm{lo}}(t) = A_\pi t + \log c_\pi - \log 5 + \log(1 - 16/N) + \log c_V - 10 \log t + \log F(s,k) \\ - \log(2C_Q) - st + \alpha s - 8(B - \sigma_Q) - \frac{4}{t^2} - 8(e^\alpha - 1).$$

With the constants of Table 2, $M_{\mathrm{lo}}(\log Z) \geq +0.654075$ for odd N and $\geq +9.517845$ for even N; hence $S(\mathcal{A}, z) > 0$ for every N with $z \geq 10^8$.

Proof. By Lemma 4.11, Theorem 4.1 applies at every $z \geq Z$ with $k = 2 + \log K_{\mathrm{an}}$. Write $N = c_\pi e^{A_\pi t}$ and $X \geq (N - 16)/5$ (Lemma 4.8). Lemmas 5.3 and 5.4 give $M(t) \geq M_{\mathrm{lo}}(t)$; the factor $\log^{-2} z$ from V contributes $-2 \log t$ and R_4 contributes $-8 \log t$. Then $M_{\mathrm{lo}}(t) = C_0 + (A_\pi - s)t - 10 \log t - 4/t^2 + \log(1 - 16/N)$ with a constant C_0, and

$$\frac{d}{dt}\Big[(A_\pi - s)t - 10 \log t - \frac{4}{t^2}\Big] = (A_\pi - s) - \frac{10}{t} + \frac{8}{t^3} > 0$$

whenever $(A_\pi - s)t > 10$, which holds for $t \geq \log Z$; the term $\log(1 - 16/N)$ is increasing in t. Hence M_{lo} is increasing on $[\log Z, \infty)$, with no grid. The value $M_{\mathrm{lo}}(\log Z)$ is evaluated in ball arithmetic. □

Table 2: Analytic range: constants and certified $M_{\mathrm{lo}}(\log Z)$. Rows marked "fixed K" replace K_{an} by the indicated constant. Negative values are not certified bounds; they say that the method fails with these parameters. The values of s and α are rounded to five decimals; the certified bounds use the double-precision values printed by `analytic.py`.

Q	Z	$K_{\mathrm{ex}}(Z) \le$	K_{an}	$c_V \ge$	s; α	$M_{\mathrm{lo}}(\log Z)$
$\{3,5\}$	10^8	1.000061	1.005973	2.068826	14.24820; 0.57719	**+0.654075**
$\{3,5\}$, even N	10^8	1.000061	1.005973	2.068826	14.24815; 0.57719	**+9.517845**
$\{3\}$	10^8	1.037250	1.043382	1.241296	14.39135; 0.58719	−2.264
$\{3\}$	10^9	1.037245	1.042087	1.242838	14.38041; 0.58643	**+0.371416**
$\{3\}$	10^{10}	1.037243	1.041163	1.243941	14.37186; 0.58583	+3.134105
$\{3,5\}$, fixed $K = 1.08997$	10^8				14.56237; 0.59900	−4.962
$\{3\}$, fixed $K = 1.097$	10^9 / 10^{10}					−3.680 / −1.458
$\{3\}$, fixed $K = 1.146$	10^{10}					−5.280

For $Q = \{3,5\}$ and $Z = 10^8$ one has $(A_\pi - s)\log Z = 32.269$ (odd) and 41.481 (even), both > 10, and $s \ge 2k + 3$ holds with room to spare. Figure 3 shows M_{lo} as a function of z.

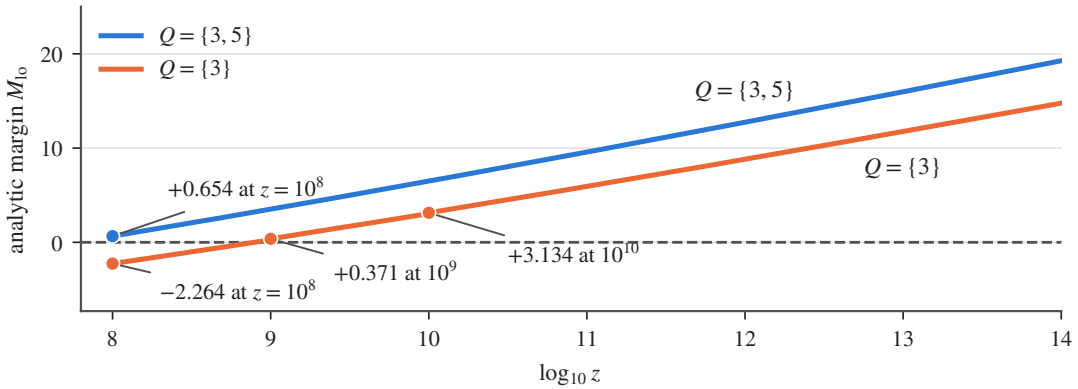


Figure 3: Analytic range: $M_{\mathrm{lo}}(t)$, $t = \log z$, with the constants and the (s, α) of Table 2 held fixed (ball-arithmetic lower endpoints; increasing in t by Proposition 5.5). Blue: $Q = \{3,5\}$, constants of $Z = 10^8$. Orange: $Q = \{3\}$, upper envelope of the curves for $Z = 10^8, 10^9, 10^{10}$ (lower bounds valid for $z \ge Z$ may be combined by maximum). The dots are the entries of Table 2; the $Q = \{3\}$ curve is negative at 10^8 and positive from 10^9.

5.4. Completion of the proof

Let $N \ge 2$. If $N \le N_0$, Corollary 3.2 applies. If $N > N_0$ is odd, put $z = (N^2/8)^{1/32} > u_0$; if $N > N_0$ is even, put $z = (N^2/4)^{1/33} > u_0^{\mathrm{e}}$. If $z \le 10^8$, z lies in a cell of Proposition 5.2 for the parity of N and Lemma 5.1 gives $S(\mathcal{A}, z) > 0$; if $z \ge 10^8$, Proposition 5.5 gives $S(\mathcal{A}, z) > 0$. Lemma 3.7 (odd N) or Lemma 3.4 (even N) then yields a, b with $N = a + b$ and $\Omega(ab) \le 32$. □

6. Remarks

6.1. A second chain with $Q = \{3\}$

The theorem also follows from a chain that uses only the constants of [7, §3] ($C_Q = 19/6$) and not Section 4.2: the finite range up to 10^{10} with minimal margin +0.079236, and the analytic range from $Z = 10^9$ with +0.371416 (Tables 2 and 3); the two ranges overlap. The theorem holds if either chain is correct; the main chain of this paper is the one with $Q = \{3,5\}$ of Section 5.

Table 3: Minimal certified margins of the finite range, log(main/error). I: per-cell K; E: suffix-maximum envelope of the per-cell K; U: uniform K. Negative U values come from a grid search and mean that the method fails, not certified upper bounds.

K	Q	up to	cells	failing	min. margin	cell	s	δ	K
I	$\{3,5\}$	10^8	117 415	0	**+1.118347**	(10342.3367, 10343.3709]	14.30443	107/128	1.002250
I	$\{3\}$	10^{10}	122 023	0	+0.079236	(15822.3670, 15823.9492]	14.44136	109/128	1.039801
I, even N	$\{3,5\}$	10^8	118 322	0	+5.024006	(2802.7202, 2803.0005]	14.34590	102/128	1.005873
E	$\{3,5\}$	10^8	117 415	0	+1.005101	same as I			1.005973
E	$\{3\}$	10^{10}	122 023	0	+0.036773	same as I			1.041163
U ($K = 1.146$)	$\{3\}$	10^8	117 415	85 342	−3.405836	$z \approx 1.18 \cdot 10^5$			
U ($K = 1.089970$)	$\{3,5\}$	10^8	117 415	71 182	−1.671836	$z \approx 4.29 \cdot 10^4$			

The envelope variant E, in which $K(z) = \sup_{z'' \geq z} K(z'')$, has the literal form "for every $z \geq z_a$" of the lemmas of [7]. It has the same logical status as the per-cell reading: both rest on Theorem 4.1.

6.2. A uniform constant does not suffice

With Campbell's uniform $K = 1.146$ and optimised s, δ, the finite range fails on 85 342 of 117 415 cells; with $Q = \{3,5\}$ and the uniform $K = 1.089970$ (the supremum over the range) it still fails (Table 3). So does the analytic range with fixed constants (Table 2). The per-cell constant, hence Theorem 4.1, is essential.

6.3. Why $\Omega(ab) \leq 31$ does not follow

Table 4 gives exploratory floating-point profiles (a grid of 60 points; these are not certified bounds) of the minimum of log(main/error) for the bound 31.

Table 4: Exploratory profiles for $\Omega(ab) \leq 31$ (floating point).

case	relation	minimum	where
odd, $Q = \{3\}$	$N = \sqrt{8}\, z^{15.5}$	−5.312	$z \approx 2.03 \cdot 10^5$; negative on $[534, 9.9 \cdot 10^7]$
odd, $Q = \{3,5\}$	same	−3.947	$z \approx 5.9 \cdot 10^4$
odd, $Q = \{3,5,7\}$	same	−4.167	pre-sieving more is worse
even, $Q = \{3\}$	$N = 2z^{16}$	−0.274	negative on $[4659, 5.6 \cdot 10^4]$

The deficit of 4 to 5 in the logarithm for odd N cannot be closed by constants; it seems to require a weighted sieve or a better factor than $F(s,k)$, as suggested in [7, §6]. Figure 4 shows the profiles for $Q = \{3\}$ and $Q = \{3,5\}$.

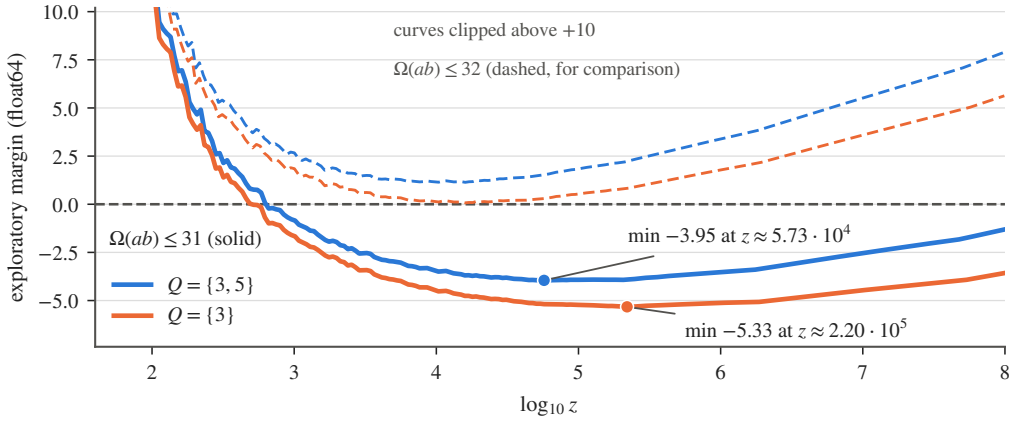


Figure 4: Why $\Omega(ab) \le 31$ does not follow. Solid: exploratory *floating-point* margin for the bound 31 ($N = \sqrt{8}\, z^{15.5}$, float64, grid of 240 points, hence slightly below the 60-point minima of Table 4; not a certified bound), negative over most of the range with minima -3.95 ($Q = \{3, 5\}$) and -5.33 ($Q = \{3\}$). Dashed: the same computation for the bound 32, which is positive (certified in Figure 1).

Appendix A. Proof of Theorem 4.1

Throughout this appendix z is fixed, $\mathcal{A}$, $\mathcal{P}$, g, X, r_d are as in Section 2, and all divisors d, e, m run over divisors of $P(z)$; they are square-free. The weights θ_e below are not the remainders ρ_ℓ of (2.3), and Y_0 is not $Y = N - 1$. With h and $I(U, z)$ as in Section 4.1, $h(p) > 0$ and $V(z)^{-1} = \prod_{p|P(z)}(1 + h(p)) = \sum_{d|P(z)} h(d)$. Let $D = z^s$ with $s \ge 2k + 3 \ge 3$ and $Y_0 = \sqrt{D/z} = z^{(s-1)/2} \ge z$.

A.1 Weights. Let $\psi(d) = \min\{1, \log(Y_0/d)/\log z\}$ for $d < Y_0$ and $\psi(d) = 0$ for $d \ge Y_0$; then $0 \le \psi \le 1$, $\psi(1) = 1$, and $\log d \mapsto \psi(d)$ is $(1/\log z)$-Lipschitz. Put $J = \sum_d h(d)\psi(d)$, so $1 \le J \le V(z)^{-1}$, $y_d = \psi(d)/J$, and define θ_e by

$$g(e)\theta_e = \mu(e) \sum_{d:\, e|d} h(d) y_d. \tag{A.1}$$

Then $\theta_1 = \sum_d h(d) y_d = 1$ and $\theta_e = 0$ for $e \ge Y_0$. Moreover $|\theta_e| \le 1$: since $h(e)/g(e) = \prod_{p|e}(1 + h(p)) = \sum_{t|e} h(t)$,

$$|\theta_e| = \frac{1}{J} \sum_{t|e} \sum_{f:\,(f,e)=1} h(t)h(f)\,\psi(ef) \le \frac{1}{J} \sum_{t|e} \sum_{f:\,(f,e)=1} h(tf)\,\psi(tf) \le 1,$$

because ψ is non-increasing, $tf \le ef$, and distinct pairs (t, f) give distinct $d = tf$. Möbius inversion of (A.1) gives

$$\sum_{m:\, e|m} g(m)\theta_m = \mu(e)h(e)y_e. \tag{A.2}$$

A.2 The lower bound sieve. For every integer a,

$$\mathbf{1}_{(a,P(z))=1} \;\ge\; \Big(1 - \sum_{p|(a,P(z))} 1\Big)\Big(\sum_{d|(a,P(z))} \theta_d\Big)^2,$$

since both sides equal 1 if $(a, P(z)) = 1$, and otherwise the first factor is ≤ 0. Summing over n gives $S(\mathcal{A}, z) \ge \sum_d \lambda_d |\mathcal{A}_d|$ with

$$\lambda_d = \lambda_d^+ - \sum_{p|d} (\lambda_d^+ + \lambda_{d/p}^+), \qquad \lambda_d^+ = \sum_{[d_1,d_2]=d} \theta_{d_1}\theta_{d_2},$$

hence $S(\mathcal{A}, z) \ge XV(D, z) - \sum_d |\lambda_d||r_d|$ with $V(D, z) = \sum_d \lambda_d g(d)$.

A.3 The main term. Since $g([d_1,d_2]) = g(d_1)g(d_2)/g((d_1,d_2))$ and $1/g(m) = \sum_{e\mid m} 1/h(e)$, (A.2) gives $\sum_{d_1,d_2}\theta_{d_1}\theta_{d_2}g([d_1,d_2]) = \sum_e h(e)y_e^2$. For a fixed p, write $d_i = p^{\varepsilon_i}d_i'$ with $p\nmid d_i'$; then $g([p,d_1,d_2]) = g(p)g([d_1',d_2'])$, and the same diagonalisation over $p\nmid e$, with $\sum_{e\mid m,\ p\nmid m} g(m)(\theta_m+\theta_{pm}) = \mu(e)h(e)(y_e - y_{pe})$ by (A.2), gives

$$V(D,z) = \sum_d h(d)y_d^2 - \sum_{p\nmid d} g(p)h(d)\big(y_d - y_{pd}\big)^2. \tag{A.3}$$

This is [8, (2.3)]. Using $g(p)\le h(p)$ and putting $m = pd$,

$$V(D,z) \ge \sum_m h(m)\Big(y_m^2 - \sum_{p\mid m}(y_{m/p}-y_m)^2\Big).$$

If $m\le Y_0/z$, then $\psi(m/p) = \psi(m) = 1$; if $m\ge Y_0z$, then $m/p > Y_0$ and $\psi(m/p) = \psi(m) = 0$; in both cases the inner sum vanishes. If $Y_0/z < m < Y_0z$, the Lipschitz property gives $\sum_{p\mid m}(y_{m/p}-y_m)^2 \le J^{-2}\sum_{p\mid m}(\log p/\log z)^2 \le J^{-2}\log m/\log z$, as $p<z$. Dropping $y_m^2\ge 0$ on this range,

$$J^2V(D,z) \;\ge\; \sum_{m\le Y_0/z} h(m) - \frac{\log(Y_0z)}{\log z}\sum_{m>Y_0/z} h(m) \;\ge\; V(z)^{-1} - \frac{\log(Y_0z^2)}{\log z}\,I(Y_0/z,z),$$

and $\log(Y_0z^2)/\log z = (s+3)/2$. (In [8, §2.3] the normalisation is written $J = \sum_{d\le Y_0}h(d)$; the argument only needs $\theta_1 = 1$ and $J\le V(z)^{-1}$, which hold for the J above.)

A.4 The remainder. For d with $\omega(d) = \omega\ge 1$, $|\lambda_d^+|\le 3^\omega$ because $|\theta|\le 1$, so $|\lambda_d|\le(\omega-1)3^\omega + \omega 3^{\omega-1} = 3^{\omega-1}(4\omega-3)\le 1.38\cdot 4^\omega\le 2\tau_4(d)$, and $\lambda_1 = 1$. If $\lambda_d\ne 0$, then $d\mid p\,d_1d_2$ with $p<z$ and $d_1,d_2<Y_0$, so $d<zY_0^2 = D$. Hence $\sum_d|\lambda_d||r_d|\le 2R_4(\mathcal{A},D)$, using $r_1 = 0$.

A.5 Conclusion. Apply Proposition 4.4 with $U = Y_0/z$, so that $u = \log U/\log z = (s-3)/2\ge k$:

$$\frac{s+3}{2}\,I(Y_0/z,z)V(z) \le \frac{s+3}{2e^k}\Big(\frac{2ek}{s-3}\Big)^{(s-3)/2} = 1-F(s,k).$$

If $F(s,k)\le 0$, the claim $S(\mathcal{A},z)\ge XV(z)F(s,k) - 2R_4(\mathcal{A},D)$ is trivial, since $S(\mathcal{A},z)\ge 0$. If $F(s,k)>0$, A.3 gives $J^2V(D,z)V(z)\ge F(s,k)>0$, and $J\le V(z)^{-1}$ gives $V(D,z)\ge F(s,k)/(J^2V(z))\ge V(z)F(s,k)$. By A.2 and A.4, $S(\mathcal{A},z)\ge XV(z)F(s,k) - 2R_4(\mathcal{A},D)$. The dimension condition was used only in Proposition 4.4, at the level z. □

Numerical check. The script `test_local_sieve.py` in the directory `certificate/local_sieve` of the repository (output file `out_test_local_sieve.txt`) tests these steps for 300 random densities and for the sequences $\mathcal{A}$ of this paper with $Q = \{3\}$ and $\{3,5\}$, odd N and even N in $[41,400)$ together with four larger values up to 9999, $z\le 18$ and $s\in[2,24]$ (29 040 cases), with no violation. With the weights $\psi(d)$ replaced by rational approximations, the identity (A.3), $\theta_1 = 1$, $|\theta_e|\le 1$, $|\lambda_d|\le 2\tau_4(d)$, the support $d<D$ and the inequality $S(\mathcal{A},z)\ge\sum_d\lambda_d|\mathcal{A}_d|$ are checked in exact rational arithmetic; the bound of A.3, Proposition 4.4 and the conclusion $S(\mathcal{A},z)\ge XV(z)F(s,k) - 2R_4(\mathcal{A},D)$ (with $\kappa = 2$ and the smallest K allowed at this z) are checked in floating point with a tolerance of 10^{-9}, relative to $V(z)^{-1}$ for the bound of A.3 and absolute for the last two, which are checked in the 20 328 cases with $s\ge 2k+3$, of which 11 181 have $F(s,k)>0$. The test checks the algebra; it is not part of the proof.

Use of AI tools in the research

Claude Opus 5.5 (Anthropic), used through the Claude Code agent environment, took part in the research as follows. It first proposed and derived the three reductions of Section 3; the adaptation to $Q = \{3,5\}$ (Lemmas 4.5–4.8); the reduction of the per-cell constant to finitely many candidates (Lemma 4.10); Lemmas 4.11–5.4 and

the monotonicity argument of Proposition 5.5; the analysis of the reading of (2.1) in Section 4.1, including the literature search behind Remark 4.3; Proposition 4.4; the proof of Theorem 4.1 in Appendix Appendix A, including the case distinction on the sign of F, which was found by a numerical test of each step (script `test_local_sieve.py`); and the treatment of even N (Lemma 3.4). It wrote all the code of the certificate (prime sieve, ball-arithmetic bounds, cell cover, analytic range) and the scripts that draw Figures 1–4 directly from the certificate outputs. The checks described under "Checks by separate implementations" were written by separate instances of the same tool, which did not import the main code; their agreement is internal consistency, not independent human verification. The checks that do not rely on AI-written arithmetic are the comparison of prime counts and block sums with primesieve 12.17 and of $\pi(10^k)$, $k \le 10$, with known values. The author checked every proof by hand; the certificate is deterministic, and re-running the published code reproduces all certificate outputs bit for bit except the wall-clock timing lines (Code availability statement).

Code availability statement

Code and certificate outputs are available at `https://github.com/giovanedms/goldbach-omega32` and archived at Zenodo, `https://doi.org/10.5281/zenodo.23456153` (all versions); the certificate described here is release v1.0.0, `https://doi.org/10.5281/zenodo.23456154`. The repository contains: `primes_blocks.py` (exact counts and sums of primes per block up to 10^{10}), `check_blocks.py` (comparison with primesieve), `rlib.py` (ball arithmetic), `run_cover.py` (finite range), `analytic.py` (analytic range), `test_local.py` (brute force of the local lemmas), `test_local_sieve.py` (numerical test of the algebra of Appendix Appendix A), and their outputs. Environment: Python 3.14.4, python-flint 0.9.0, numpy 2.5.4, mpmath 1.3.0, primesieve 12.17; no GPU. The main finite cover (`run_cover.py 3,5 1e8 interval`) runs in about 40 s; the even case is the same command with the environment variable `GB_PARITY=even` (about 70 s), followed by `analytic.py 3,5 1e8`. Campbell's scripts [12] were re-run at commit `2755a58b6eb1...` and reproduce the versioned output byte for byte. SHA-256 of the main files:

```
7e8eb38ada4507a5599631cd2511db23d1d81d9f501a29d8c5da5e8163b1566c  out_cover_Q35_interval.txt
c1d879bed47e8b3ce647e62bb1b9657529ea7d96f051cb34b7745a221240bad8  out_cover_Q35_interval_even.txt
9d4dba8e248b89e284dbb3e57e16f2474b03b72bbb8725c30db6fa90e677259b  out_cover_Q3_interval.txt
b0d1624d1eb576a1ab9066db8abb06c54954598467d58a5e49382d23808bb0fa  out_analytic.txt
ca269994cbf54fbb438422fd18c38bd1aad9e2f10dce3f940375fc64a3cdbe91  out_analytic_even.txt
f44c4b01fddeb9b4b6e933fe291daedf3b84247dce68b24abe6490bce1c44b9a  rlib.py
0acb487e5dd3bffef14ce2c5a9b358d41cf19b06e53015349af49c713f236c5a  run_cover.py
84483ca099c5029ab112e445ab48e26542f2cf9af4f2770745d56628c7732356  analytic.py
4e901d77dfc171bce886f9924b9d8a0e59eb26b0b517a7ec53198cb7c552d86f  primes_blocks.py
5efb2514a9e8bec4db66844a00723fb703ac50ed3d973475fa98c4966336151d  test_local_sieve.py
5e21da3a192ba96e952c4e29602894e695703b6a7c49a7526e93fd5c97457ee1  out_test_local_sieve.txt
```


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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude Opus 5.5 (Anthropic), through the Claude Code agent environment, to draft the manuscript from a technical dossier of the proofs, and to organise and edit the text. The use of the same tool in the research itself is described in the section "Use of AI tools in the research" above. After using this tool, the author reviewed and edited the content as needed, checked the proofs by hand, and takes full responsibility for the content of the published article.

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Declaration of competing interest

The author declares no competing interests.

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